THE OPEN PROBLEMS OF GROTHENDIECK'S 1973 BUFFALO LECTURES, AND WHAT BECAME OF THEM

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Abstract. In the spring and summer of 1973 Grothendieck gave three courses at the State University of New York at Buffalo, on topos theory, on algebraic geometry and on group schemes, whose recordings are online. In them he asks some twenty questions he cannot answer, states conjectures, and hands problems to the audience. We list them from the recordings, with the second at which each is asked, and say what the literature has made of each. Most are settled. Several went against what he expected: his characterisation of presheaf topoi, whose two conditions instead characterise the essential subtopoi of presheaf topoi, the non-existence of a right adjoint $f^!$, the colimits of accessible categories, the descent of immersions that are not quasi-compact, and his doubt that quasi-coherent sheaves can be pulled back fully faithfully along universal effective epimorphisms. Perrin proved his main conjecture on group schemes in 1976, and the effectivity of finite equivalence relations had been answered by Venken in 1971, two years before he asked it. Some answers are ours; their proofs are given in an appendix. What still looks open concerns large classes of limits in his « problem three », one question that depends on set theory, and descent along non-quasi-compact universal effective epimorphisms in ZFC.

1. The lectures and the method

The recordings are on a YouTube channel under the pseudonym « Erwin Schulhoff »: four videos for the topos course (April–July 1973), four for algebraic geometry (May–July) and two for group schemes (June–July), the last two also carrying some sessions of the algebraic geometry course. The edition has made an index of results for each course, with the second at which each statement is made, and reconstructed the blackboards [35]. For this note we searched the transcripts for the places where he says that something is not known, that he cannot prove it, that he expects or suspects it, or where he calls it a conjecture or a problem; we kept the ones he frames as questions, and dropped asides and routine omissions. Each problem below gives the video and the time (a link to that second), how he frames it, and its status, with the references we checked. Where we give an argument of our own, not found in the literature, it is marked $^{(e)}$.

The transcripts are first passes from captions and Whisper that no one has checked, and a few exchanges with the room are hard to attribute. He also corrected himself: on 10 July he retracted several claims of 2 July on algebraic theories (the transitivity of strict cogeneration, a two-step construction, a criterion by strict cogenerators). We report the question as finally posed.

2. Topoi

Video 2 of the topos course is `PqCQM3Pcl6Q` unless stated.

Problem 1 (1:26:22 and 7:31:23)**.** *Are the presheaf topoi $\widehat{C}$ exactly the topoi that share all the exactness properties of* **Set***, including those involving infinite limits: arbitrary sums and products distributive, and arbitrary products of epimorphisms epimorphic?*

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He suspects it, says it should not be too hard, and returns to it later as a characterisation of what he calls essential topoi, at least when they have generators.

Status. False, and the class his two conditions define can be described exactly[(e)]. Write (A) for « arbitrary products of epimorphisms are epimorphisms » and (B) for « arbitrary products distribute over arbitrary sums ». For a Grothendieck topos E, (A) and (B) together are equivalent to each of: E is an essential subtopos of a presheaf topos; for a small generating subcategory with its induced topology, covering sieves are closed under arbitrary intersections; E is totally distributive in the sense of Rosebrugh and Wood. The last equivalence is Lucyshyn-Wright's theorem [23, 1.5.5]; the others are proved in Appendix A, using the axiom of choice. Condition (B) alone says that E is locally connected, and the two conditions are independent: sheaves on the Cantor set satisfy (A) and not (B), sheaves on $\mathbf{R}$ satisfy (B) and not (A). A topos with both and not a presheaf topos is the topos of sheaves on $[0, 1)$ whose open sets are the intervals $[0, a)$: its frame is the complete chain $[0, 1]$, which is completely distributive, so (A) and (B) hold, but it has no nonzero projective object, while a presheaf topos is generated by its projective representables; the presheaf topoi are exactly the topoi with (B) and enough projectives (Proposition A.11). In the localic case, $\mathrm{Sh}(L)$ satisfies (A) and (B) exactly when L is completely distributive, and is a presheaf topos exactly when L is join-generated by completely join-prime elements. His « essential » is not the essential geometric morphism of later usage; that the class he named turns out to be that of the essential subtopoi of presheaf topoi looks like a coincidence of words. We do not claim that these equivalences are new, only that we have not found them stated.

Problem 2 (4:05:50)**.** *For a morphism of topoi f, can f_* have a right adjoint $f^!$ on sheaves of sets without f being an equivalence?*

He has « a feeling » that it cannot, and concludes that the notation $f^!$ has no reason to exist for topoi.

Status. False, even for an open dense immersion, the case he was testing. For the open point j of the Sierpiński space S, whose sheaves are the maps $A \to B$ (sections over S, then over the open point), j_* is the diagonal $A \mapsto (A \to A)$, which is the constant-sheaf functor of S, and its right adjoint is the global-sections functor $\Gamma^{(e)}$. More generally, for a small category C with a terminal object the global-sections functor $\Gamma\colon \widehat{C} \to \mathbf{Set}$, the direct image of $\widehat{C} \to \mathbf{Set}$, has a right adjoint: $\widehat{C}$ is a local topos, and such morphisms are the local maps studied by Johnstone and Moerdijk [18].

Problem 3 (5:18:59)**.** *Every type of algebraic structure, in a broad sense, should have a classifying topos with a universal model, added axioms should correspond to subtopoi, and the homotopy type of the classifying topos should be a classifying space of the structure.*

He states it as a programme and says he has checked it on a dozen or two examples.

Status. Carried out for geometric theories: every geometric theory has a classifying topos with a generic model (Joyal–Reyes; Makkai–Reyes [25]; Hakim for rings and local rings [15]), quotient theories correspond to subtopoi (Caramello [5]), and for groups and categories the classifying topos has the weak homotopy type of the classifying space (Moerdijk [29]). The restriction is that « structure » means geometric theory.

Problem 4 (8:38:47, 8:46:26)**.** *Can every exactness property be written as « every D-functor $I \to C$ factors through a given D-functor $I \to I'$ »? Is « every semigroup object is a monoid » an exactness property?*

This comes from a discussion with the room after the session, and who says what is not always clear.

Status. Formalised by Jacqmin and Janelidze as exactness sequents of sketches, which cover what they call first-order exactness properties, among them properties of internal structures

such as an internal monoid being a group, and not higher-order ones such as having enough projectives [16]. « Every semigroup object is a monoid » holds only in the trivial category: the left-zero multiplication $X \times X \to X$ is associative, and a unit $e\colon 1 \to X$ for it forces $\mathrm{id}_X = e \circ !$, so $X \cong 1$; since every object carries this multiplication, the category is trivial$^{(e)}$.

Two further questions are answered on the spot. Asked for a criterion characterising sober spaces (1:55:37), he says he knows none that can be read off the topos, since a space and its sobrification have the same sheaves; the intrinsic criteria are standard [19]. And the universal problem for torsors under a group G of a topos S (`ORG14LNb5gE`, 3:15:58) he solves at once with the topos of G-objects of S, as in Giraud [14].

3. Algebraic theories relative to a class of limits

For a class D of limits and colimits, he considers the 2-category $\mathbf{Cat}_D$ of categories with D-limits and D-functors, and calls a theory algebraic if it is obtained from a set I of base objects by stepwise rules: the free D-envelope, adjoining arrows, imposing axioms.

Problem 5 (8:34:57)**.** *Are the algebraic theories relative to D exactly the theories $C \mapsto \mathrm{Hom}_D(R, C)$ 2-represented by objects R of $\mathbf{Cat}_D$?*

Status. Settled: yes for small R, which is his own argument and the modern presentation of small categories with limits by generators and relations (sketches, essentially algebraic theories [2, ch. 3]); no in general. For D the small limits, $R = \mathbf{Set}$ is representable, with $\mathbf{Set}^{\mathrm{op}}$ as its category of models in $\mathbf{Set}$, which is not accessible, while stepwise theories have locally presentable categories of models; he gives this example himself on 10 July. For D all small limits the algebraic R are, up to duality, the opposites of locally presentable categories, the form Gabriel–Ulmer duality gives [13].

Problem 6 (10:03:06)**.** *For which D are the envelope problem (the free D-completion of a category) and the localisation problem in $\mathbf{Cat}_D$ solvable? He knows them for finite limits and for topoi, and not for additive, abelian or Karoubian categories.*

Status. Settled for every small class D by Kelly's transfinite constructions and two-dimensional monad theory [21, 4], with explicit answers: the idempotent completion for Karoubian categories, the biproduct completion of the free preadditive category for additive ones, the free abelian category of Freyd and Adelman [12, 1], the exact completion of Carboni and Celia Magno [6]. For a large class, such as all small limits, the answers depend on size, as he noted on 10 July: localisations of large categories need not be locally small.

Problem 7 (`4gynb9uTEE4`, 2:03:54 and 2:10:23)**.** *« Problem three »: is $I \to R$ basic if and only if every strictly full subcategory of R stable under D-limits and containing the image of I is R itself? And, conjecturally, can I be enlarged so that no arrows remain to be adjoined, only axioms?*

Status. Settled, except for large classes $D^{(e)}$. Write (G) for the condition of the problem; it holds exactly when the transfinite closure of I under D-limits exhausts R, after ρ steps when the shapes of D have fewer than ρ arrows [20, §3.5]. Assuming, as he does, that envelopes, localisations and the other constructions exist in $\mathbf{Cat}_D$, Appendix B proves that basic implies (G) for every D (Theorem B.7), and that $I \to R$ is basic if and only if (G) holds and R is algebraic over some base (Theorem B.9). For a small class D every R satisfying (G) is essentially small, hence algebraic, so the equivalence holds with the base fixed. For D all small limits it fails: every set is an equaliser of two maps between powers of 2, so $\{2\} \to \mathbf{Set}$ satisfies (G) after two steps, but $\mathbf{Set}$ is not algebraic, since 2 is not presentable in $\mathbf{Set}^{\mathrm{op}}$ (§B.5). The second question has a positive answer in the locally presentable case by Gabriel–Ulmer: taking for I the opposite of the λ-presentable objects, the category of models is the category of λ-continuous functors on I, so that only axioms remain to be imposed [13]. What remains open is the converse for large classes beyond these cases, such as finite limits with all colimits but no exactness condition, and whether the number of closure steps is bounded by a small

ordinal when the shapes of D are unbounded. Two related statements of the course on topoi (`ORG14LNb5gE`), that every object of a later stage is covered by images of earlier ones and that the finite products of a basic I generate, are false as stated: adjoining an arrow $1 \to 1 \sqcup 1$ to **Set** gives **Set** $\times$ **Set**, in which $(1, 0)$ is not covered by the diagonal (§B.7).

Around these he asks whether faithfulness of $\mathrm{Hom}_D(R, C) \to C^I$ for all C suffices for algebraicity (9:40:36, 10:00:19). It does not, as he reports himself later, citing an example he attributes to Isbell (the name is uncertain in the recording; 2:03:54). In the session itself he and the room arrive at lattices, and a large ordinal; **Set** with $I = \{2\}$ is another counterexample, faithfulness holding because every set is a limit of powers of $2^{(e)}$. His hope of a bound on the number of limit operations (8:08:14) he retracted himself on 10 July.

4. Accessible categories

Problem 8 (`4gynb9uTEE4`, 1:31:03)**.** *Is a small 2-colimit in* **Cat** *of accessible categories accessible?*

Status. No in general$^{(e)}$. Let B be the free category on arrows $a\colon u \to v$, $b\colon v \to w$, $h\colon u \to w$, map it to the chain $0 < 1 < 2$ and to the free monoid on one generator t, sending a, b, h to t. The pushout, strict or pseudo, is the one-object category with endomorphisms $\{1, t\}$ and $t = t^2$, since $h = ba$ in the chain; the idempotent t does not split, while accessible categories split idempotents. The three categories are small and idempotent-complete, hence accessible, and the functors are accessible. He expected the answer yes, while noting that a sum of two categories with colimits has none. Directed colimits of accessible categories along accessible full embeddings are accessible (Paré–Rosický [32]). The limit side, which he recalls from SGA 4, is right in modern form [24, ch. 5].

Problem 9 (`4gynb9uTEE4`, 1:51:31)**.** *Is a full subcategory of an accessible category, stable under limits and with a left adjoint inclusion, accessible?*

Status. Yes under an extra hypothesis; in ZFC alone we have not found it settled. When the inclusion is also accessible, the condition he proposed adding, the answer is yes in ZFC: for a locally presentable ambient category this is [2, 2.78]. Under Vopěnka's principle the answer is yes for a locally presentable ambient category [2, ch. 6]. Under the weak Vopěnka principle every limit-closed full subcategory of a locally presentable category is reflective, and this fails in the category of graphs when the weak principle fails [3], [2, 6.22]; Wilson showed that the weak principle does not imply the full one [39]. A reflective, limit-closed, non-accessible subcategory in a model of the weak principle without the full one would show that ZFC does not decide the question; we have not found one in print.

Problem 10 (`4gynb9uTEE4`, 4:48:17)**.** *If E and E^{op} are both accessible, is E equivalent to the point?*

Status. No: the two-element chain is a counterexample, being, like its opposite, small and idempotent-complete. With small colimits in E, as he assumes for his main theorem, E and E^{op} are both locally presentable, and then E is a preorder, hence a complete lattice (Gabriel–Ulmer [13, Satz 7.13]).

5. Algebraic geometry and group schemes

Problem 11 (`vuys5SNs2eQ`, 3:31:11; see also 2:59:31)**.** *Do immersions that are not quasi-compact descend along a faithfully flat quasi-compact morphism? At 2:59:31: is descent of subschemes exact without retrocompactness? (« you asked for an open problem »).*

He believes the first but cannot prove it; his attempt by a filtered union of quasi-compact opens fails on his own example, the closed points of the line.

Status. Both are false. Open, closed and quasi-compact immersions descend along fpqc coverings, and immersions along fppf ones [38, 02L3, 02L6, 02L8, 02YM]; on the underlying sets, locally closed and retrocompact subsets descend along fpqc morphisms, and locally closed ones along fppf morphisms [37, Thms 4.1, 4.2]. The disjoint union of the $\operatorname{Spec}\mathbf{F}_p$ mapped to $\operatorname{Spec}\mathbf{Z}$ is not an immersion, but becomes one after the faithfully flat base change $\operatorname{Spec}\prod_p \mathbf{Z}_{(p)} \to \operatorname{Spec}\mathbf{Z}$; the same subscheme answers the second question. The example, base change included ($\prod_q \mathbf{Z}_{(q)}/p \cong \mathbf{F}_p$), is in a comment of August 2026 on Stacks tag 02YL, which we checked; it is not refereed, and we found no earlier published counterexample. The subscheme is the one he used in class, the closed points, taken after base change.

Problem 12 (`3ULFIl_2Wis`, 5:04:48)**.** *« The main conjecture »: a connected separable group scheme over a field is an extension of a group of finite type by an affine group, the quotient map faithfully flat and quasi-compact; equivalently a filtered limit of groups of finite type with affine transition maps.*

Status. Proved by Perrin [34], a summary of his Orsay work: every quasi-compact group scheme over a field is a filtered limit of algebraic quotients with affine kernels, with transition maps affine from some index on; he notes that the statement had been conjectured in EGA IV. A connected group scheme over a field is quasi-compact, as Grothendieck recalls just after the statement. Perrin's counterexample with abelian varieties of growing dimension is the phenomenon Grothendieck wanted to exclude at 5:11:21, the dimensions of the abelian quotients being bounded.

Asked for a characterisation of schemes among algebraic spaces (6:39:12, a session of the algebraic geometry course recorded on this video), he answers that it would solve the representability problem, and no; there are sufficient criteria and partial results [38, 03XX, 0ADD, 03JI, 0D2W]. Smaller questions are settled. In the algebraic geometry course: whether a Noetherian spectrum, the ascending chain condition on radical ideals, is the same as the ascending chain condition on primes (`51wmmpHand0`, 7:42:27; no: an infinite product of fields has only maximal primes and an infinite, non-Noetherian spectrum$^{(e)}$; see Ohm and Pendleton for rings with Noetherian spectrum [30]); and whether a scheme can have no closed point (`Oqhr-Gt66es`, 0:10:49; yes [38, 01IY]). There he also states that the local schemes are those with a unique closed point. That holds for quasi-compact schemes, but not in general: $W \sqcup \operatorname{Spec} k$, with W a scheme without closed point, has a unique closed point and is not local$^{(e)}$. In the group-schemes course: whether a monoid scheme with a closed unit section is separated (`3ULFIl_2Wis`, 3:56:26; no: adjoin a unit, as an extra rational point, to a non-separated scheme with a rational point c and the constant multiplication $xy = c^{(e)}$).

Problem 13 (`3ULFIl_2Wis`, 6:26:20–6:27:24)**.** *Along a morphism of universal one-descent, that is a universal effective epimorphism of schemes, do quasi-coherent sheaves descend? Is pulling them back at least fully faithful?*

A listener asks why the topology is not defined by all such morphisms; he answers that it would be the finest one for which representable functors are sheaves, but that these morphisms are hard to handle, that one does not know whether quasi-coherent sheaves descend along them or whether pulling back is even faithful, and that he doubts this can be proved. This is an algebraic geometry session recorded on the group-schemes video.

Status. Mostly settled$^{(e)}$, and partly against his doubt (Appendix C). Pulling back quasi-coherent sheaves along any universal effective epimorphism is fully faithful, with no quasi-compactness: test against the square-zero extension $\mathcal{O}_Y \oplus F$ and the affine line. Along a quasi-compact one, descent is effective; this is stated by Mesablishvili [26], and Theorem C.4 gives an independent proof through a finite affine cover, which reduces to the affine case, where universal effective epimorphisms are the pure ring maps and descent holds exactly along pure maps [38, 0EUA, 08XA], [28]. Without quasi-compactness, effectivity can fail: if some nonprincipal ultrafilter U on the primes is generated by $\aleph_1$ sets, for instance under the

continuum hypothesis, the family $\{\operatorname{Spec}\mathbf{Z}_A \to \operatorname{Spec}\mathbf{Z}\}_{A\in U}$ of the Stacks Project [38, 0EUG] is a universal effective epimorphism along which some descent datum, an extension of $\mathcal{O}$ by $\bigoplus_p \mathbf{Z}/p$, is not effective; the obstruction is a clopen set of $\beta\mathbf{N}\setminus\mathbf{N}$ minus the point U that does not extend, which exists at points of character $\aleph_1$ by the butterfly lemma of Fine and Gillman [31]. In ZFC alone the question is open: in the model of van Douwen, Kunen and van Mill the obstruction used here vanishes.

6. Effectivity

At the end of the algebraic geometry session of 11 July, recorded on the second group-schemes video (`3ULFIl_2Wis`, 6:59:50), asked for open problems, he gives the effectivity of finite equivalence relations on a finite scheme over a field, posed in FGA in 1961, and says he suspects a negative answer. It had been answered negatively, over every field, by Venken in 1971: Corollary 3.1 of a note on the descent of flat modules gives a non-effective equivalence relation on the spectrum of a local Artin algebra of dimension 33 [7]. A companion note checks that example by machine and gives a second one, of length 15 (16 in characteristic 2), by restricting Raicu's relation on the plane [8].

7. Summary

Problem	His stance	Status
1 presheaf topoi	suspects yes	false: Scott sheaves on $[0,1]^{(e)}$; essential subtopoi of $\widehat{C}$ [23]
2 $f^!$	feels no	false; local morphisms [18]
3 classifying topoi	programme	done for geometric theories
4 exactness properties	open in discussion	formalised [16]; semigroup case trivial$^{(e)}$
5 algebraic = representable	aim	yes for small R, no in general
6 envelopes, localisations	known in two cases	small D: yes [21, 4]
7 problem three	neither direction proved	$\Rightarrow$ yes; $\Leftrightarrow$ for small D; fails for all limits$^{(e)}$
8 colimits of accessible	expects yes	no$^{(e)}$; directed, full embeddings: yes [32]
9 reflective subcategories	asks for a counterexample	yes if the inclusion is accessible, or under Vopěnka [2]; ZFC alone open to us
10 E, E^{op} accessible	guesses the point	no: the two-element chain [13]
11 descent of immersions	believes yes	no (Stacks comment, 2026)
12 structure of group schemes	main conjecture	Perrin 1976 [34]
13 descent of quasi-coherent sheaves	doubts it	fully faithful; effective if quasi-compact; can fail under CH$^{(e)}$
§6 effectivity	suspects no	no, over every field (Venken 1971) [7, 8]

The questions that still look open are the converse of problem three for large classes D and the bound on its number of steps (Problem 7), the status in ZFC alone of Problem 9, and the effectivity of descent of quasi-coherent sheaves along non-quasi-compact universal effective epimorphisms in ZFC (Problem 13). Six of his expectations turned out wrong (Problems 1, 2,

8, 10, 11, and his doubt in Problem 13 about full faithfulness); his programme of classifying topoi, his main conjecture and his doubt about effectivity were borne out.

Use of language models. The questions were located in the transcripts, and the literature searched, with language models (Claude Opus 5.5, Anthropic), which also wrote a first draft of this text and the arguments marked (e), under the author's direction. The references were opened on their pages; the arguments marked (e) have not been checked by a specialist.

Appendix A. Presheaf topoi and the conditions (A) and (B)

This appendix proves the claims made under Problem 1. Throughout, E is a Grothendieck topos, the metatheory is ZFC, and the axiom of choice is used freely. The two conditions are:

(A) for every small family of epimorphisms $f_i\colon X_i \to Y_i$, the product $\prod_i f_i\colon \prod_i X_i \to \prod_i Y_i$ is an epimorphism;

(B) for every small family of small families $(X_{ij})_{i\in I,\, j\in J_i}$, the canonical map

$$\coprod_{\varphi\in\prod_i J_i} \prod_i X_{i\varphi(i)} \longrightarrow \prod_i \coprod_{j\in J_i} X_{ij}$$

is an isomorphism.

Following Rosebrugh and Wood [36], as recalled by Lucyshyn-Wright [23, §1], a locally small category E is *totally distributive* if there is a chain of adjoint functors $t \dashv c \dashv y\colon E \to \widehat{E} = [E^{\mathrm{op}}, \mathbf{Set}]$ with y the Yoneda embedding, that is, if $c = \mathrm{colim}$ has a left adjoint. An *essential subtopos* of a topos F is a topos E with a geometric inclusion $i\colon E \to F$ whose inverse image i^* has a left adjoint [23, §1.5]. Lucyshyn-Wright's Theorem 1.5.5 [23] states that the totally distributive categories having a small dense generator are exactly the essential subtopoi of presheaf topoi $\widehat{C}$, C small; every Grothendieck topos has a small dense generator, so the theorem applies to all of them. By a theorem of Kelly and Lawvere [22], recalled in [23, 1.5.6], the essential subtopoi of $\widehat{C}$ correspond bijectively to the idempotent ideals of arrows of C.

A.1. The class defined by (A) and (B).

Proposition A.1. *E satisfies (B) if and only if E is locally connected.*

Proof. Assume (B). With all $X_{ij} = 1$, condition (B) says that the constant-sheaf functor $\Delta\colon \mathbf{Set} \to E$ preserves small products. Being left exact, Δ then preserves all small limits; it also preserves all colimits, so it is accessible. Since $\mathbf{Set}$ and E are locally presentable, the adjoint functor theorem [2, 1.66] gives a left adjoint $\pi_0 \dashv \Delta$. Over $\mathbf{Set}$ such a left adjoint is automatically $\mathbf{Set}$-indexed: $X \times \Delta I = \coprod_I X$, so $\pi_0(X \times \Delta I) = \pi_0 X \times I$. Hence E is locally connected.

Conversely, a locally connected topos has a generating set of connected objects, the components of the objects of a generating set. For a connected object Q, the functor $\mathrm{Hom}(Q, -)$ preserves products, and it preserves coproducts: a map $Q \to \coprod_j X_j$ pulls the summands back to a decomposition $Q = \coprod_j Q_j$, and exactly one Q_j is nonzero, since $Q \neq 0$ and $Q = Q_{j_0} \sqcup \coprod_{j\neq j_0} Q_j$. So $\mathrm{Hom}(Q, -)$ sends the canonical map of (B) to the canonical bijection

$$\coprod_\varphi \prod_i \mathrm{Hom}(Q, X_{i\varphi(i)}) \cong \prod_i \coprod_j \mathrm{Hom}(Q, X_{ij})$$

of $\mathbf{Set}$, which uses the axiom of choice. A map of E that is bijective on $\mathrm{Hom}(Q, -)$ for every Q in a generating set is an isomorphism: it is a monomorphism because the generators separate maps, and an epimorphism because every map from a generator to its target lifts. □

Proposition A.2. *An essential subtopos of a presheaf topos satisfies (A) and (B).*

Proof. Let $i\colon E \hookrightarrow \widehat{C}$ be the inclusion, with $i_! \dashv i^* \dashv i_*$. Since i_* is fully faithful, so is $i_!$, and $\mathrm{id} \cong i^* i_!$.

(A): each $i_! f_i$ is an epimorphism, as the image of one under a left adjoint. The product $\prod_i i_! f_i$ is an epimorphism of $\widehat{C}$, since products are computed pointwise and the axiom of choice holds in $\mathbf{Set}$. The functor i^* preserves products and epimorphisms, so $\prod_i f_i \cong i^*(\prod_i i_! f_i)$ is an epimorphism.

(B): naturally, and compatibly with the canonical maps,

$$\textstyle\prod_i \coprod_j X_{ij} \cong i^*(\prod_i \coprod_j i_! X_{ij}) \cong i^*(\coprod_\varphi \prod_i i_! X_{i\varphi(i)}) \cong \coprod_\varphi \prod_i X_{i\varphi(i)}.$$ □

Proposition A.3. *A totally distributive Grothendieck topos satisfies (A) and (B).*

This also follows from Proposition A.2 and Lucyshyn-Wright's theorem; the proof below does not use the theorem.

Proof. Suppose that $c = \operatorname{colim}\colon \widehat{E} \to E$ has a left adjoint, so that c preserves all limits that exist in $\widehat{E}$; products in $\widehat{E}$ are pointwise.

(B): in $\widehat{E}$, $\prod_i \coprod_j yX_{ij} \cong \coprod_\varphi \prod_i yX_{i\varphi(i)} = \coprod_\varphi y(\prod_i X_{i\varphi(i)})$, by pointwise distributivity and because y preserves limits. Applying c, which preserves products and coproducts and satisfies $c \circ y \cong \mathrm{id}$, gives (B).

(A): let $P_i \subseteq yY_i$ be the sieve generated by f_i. In a topos the colimit of a sieve on Y is the union of the images of its members, because the equivalence relation generated by the diagram contains the kernel pair, sieves being closed under precomposition. So $c(P_i) = \operatorname{im} f_i = Y_i$. By the axiom of choice, $\prod_i P_i \subseteq y(\prod_i Y_i)$ is the sieve generated by $\prod_i f_i$. Hence $\operatorname{im}(\prod_i f_i) = c(\prod_i P_i) = \prod_i c(P_i) = \prod_i Y_i$. □

Lemma A.4. *Let (C, J) be a small site, with associated sheaf functor $a\colon \widehat{C} \to \mathrm{Sh}(C, J)$. If each $J(c)$ is closed under arbitrary small intersections, then a preserves all small limits; hence it has a left adjoint, and $\mathrm{Sh}(C, J) \hookrightarrow \widehat{C}$ is an essential subtopos. Conversely, if this inclusion is essential, each $J(c)$ is closed under arbitrary intersections.*

Proof. Write $F^+(c) = \operatorname{colim}_{S \in J(c)} \mathrm{Match}(S, F)$; the colimit is directed, since $J(c)$ is closed under finite intersections. Since $\mathrm{Match}(S, -) = \mathrm{Hom}(S, -)$ preserves products, there is a canonical map

$$\textstyle\big(\prod_i F_i\big)^+(c) = \operatorname{colim}_S \prod_i \mathrm{Match}(S, F_i) \longrightarrow \prod_i \operatorname{colim}_S \mathrm{Match}(S, F_i).$$

It is surjective: represent each ξ_i on some S_i, using the axiom of choice to choose the representatives, and restrict them all to $T = \bigcap_i S_i \in J(c)$. It is injective: if two families agree in each factor, they agree on some S_i for each i, hence on $\bigcap_i S_i \in J(c)$. So $(-)^+$ preserves products, and so does $a = (-)^{++}$. Being left exact, a preserves all small limits. It is a left adjoint between locally presentable categories, hence accessible, and the adjoint functor theorem [2, 1.66] gives a left adjoint.

Conversely, suppose that a preserves limits, and let S_i be covering sieves on c, so that each $a(S_i) \to a(yc)$ is an isomorphism. Then $a(\bigcap_i S_i)$, which is the wide pullback of the $a(S_i)$ over $a(yc)$, maps isomorphically to $a(yc)$, and $\bigcap_i S_i$ is covering. □

Lemma A.5. *Let $C \subseteq E$ be a small generating full subcategory with its induced topology, in which a sieve on c covers if and only if its members are jointly epimorphic in E. If E satisfies (A) and (B), each $J(c)$ is closed under arbitrary intersections.*

Proof. Let $(S_i)_{i \in I}$ be covering sieves on c, so that each $e_i\colon \coprod_{(d,s) \in S_i} d \to c$ is an epimorphism.

(1) By (A), $\prod_i e_i$ is an epimorphism onto c^I.
(2) By (B), its domain is $\coprod_{\varphi \in \prod_i S_i} \prod_i d_{\varphi(i)}$, and on the summand φ the map is $\prod_i s_{\varphi(i)}$.
(3) Pull back along the diagonal $c \to c^I$. Epimorphisms and coproducts are stable under pullback in a topos, so $\coprod_\varphi P_\varphi \to c$ is an epimorphism, where P_φ is the wide pullback over c of the maps $s_{\varphi(i)}\colon d_{\varphi(i)} \to c$.
(4) For each i, the map $P_\varphi \to c$ factors through $s_{\varphi(i)} \in S_i$.
(5) Cover each P_φ by objects of C. The composites $e \to P_\varphi \to c$ lie in every S_i, hence in $\bigcap_i S_i$, and they are jointly epimorphic. So $\bigcap_i S_i$ covers. □

Theorem A.6. *For a Grothendieck topos E the following are equivalent:*

(i) *E satisfies (A) and (B);*

(ii) *for some (equivalently, every) small generating full subcategory C of E, the induced topology has each $J(c)$ closed under arbitrary intersections;*
(iii) *E is an essential subtopos of a presheaf topos;*
(iv) *E is totally distributive.*

Proof. (i)⇒(ii) for every C is Lemma A.5. (ii)⇒(iii): by the comparison lemma $E \simeq \mathrm{Sh}(C, J)$; apply Lemma A.4. (iii)⇒(i) is Proposition A.2. (iii)⇔(iv) is Lucyshyn-Wright's Theorem 1.5.5 [23]; the implication (iv)⇒(i) is also Proposition A.3. □

The equivalence of (i), (ii) and (iii) rests only on the axiom of choice and standard facts (the comparison lemma, $a = (-)^{++}$, the adjoint functor theorem for locally presentable categories). We have not found (i)⇔(iii) or (i)⇔(iv) stated; Lemma A.4 is of the kind Kelly and Lawvere's paper [22] may contain, and we do not claim novelty.

A.2. **The localic case.** Let L be a frame, with the canonical topology, in which covers are joins: a down-set S of u covers u if $\bigvee S = u$. Recall that L is *completely distributive* if $\bigwedge_i \bigvee_j f_{ij} = \bigvee_{\varphi \in \prod_i J_i} \bigwedge_i f_{i\varphi(i)}$ for every family (f_{ij}); under the axiom of choice this is the same as constructive complete distributivity, as recalled in [23, §1.1] from Fawcett and Wood [10]. The inequality $\geq$ always holds.

Lemma A.7. *L is completely distributive if and only if, for every $u \in L$, the covering down-sets of u are closed under arbitrary intersections.*

Proof. Suppose L completely distributive, and let S_i be down-sets with $u \leq \bigvee S_i$ for each i. For $\varphi \in \prod_i S_i$, the element $\bigwedge_i \varphi(i)$ lies below $\varphi(i) \in S_i$ for every i, so it lies in $\bigcap_i S_i$. Hence $u \leq \bigwedge_i \bigvee S_i = \bigvee_\varphi \bigwedge_i \varphi(i) \leq \bigvee \bigcap_i S_i$.

Conversely, let (f_{ij}) be a family and $u = \bigwedge_i \bigvee_j f_{ij}$. By frame distributivity $u = \bigvee_j (u \wedge f_{ij})$ for each i, so the down-set S_i generated by the $u \wedge f_{ij}$, $j \in J_i$, covers u. By hypothesis $\bigcap_i S_i$ covers u. Each $w \in \bigcap_i S_i$ satisfies, for each i, $w \leq u \wedge f_{ij}$ for some j; choosing such a $j = \varphi(i)$ gives $w \leq \bigwedge_i f_{i\varphi(i)}$. Hence $u = \bigvee \bigcap_i S_i \leq \bigvee_\varphi \bigwedge_i f_{i\varphi(i)}$. □

The two directions can also be checked on (A) and (B) directly.

Complete distributivity implies (A) for $\mathrm{Sh}(L)$. Let $f_i \colon X_i \to Y_i$ be epimorphisms and $s = (s_i)$ a section of $\prod_i Y_i$ over u. Each s_i lifts to X_i locally, on a cover $\{u_{ik}\}_k$ of u. By complete distributivity $u = \bigwedge_i \bigvee_k u_{ik} = \bigvee_\varphi \bigwedge_i u_{i\varphi(i)}$, and over each $\bigwedge_i u_{i\varphi(i)}$ every s_i lifts, by restriction of its lift over $u_{i\varphi(i)}$, so that s lifts to $\prod_i X_i$. These elements cover u, so $\prod_i f_i$ is an epimorphism.

(A) and (B) imply complete distributivity. Let $u = \bigvee_k u_{ik}$ for each i, with $u_{ik} \leq u$. Apply (A) to the epimorphisms $\coprod_k y(u_{ik}) \to y(u)$, whose product has codomain $y(u)$. The identity section over u lifts locally: u is covered by elements v such that, for each i, v has a disjoint decomposition $v = \bigsqcup_k v_{ik}$ with $v_{ik} \leq u_{ik}$. Condition (B) applied to the subterminals $y(v_{ik})$ gives $y(v) = \prod_i \coprod_k y(v_{ik}) \cong \coprod_\varphi y(\bigwedge_i v_{i\varphi(i)})$, so $v = \bigvee_\varphi \bigwedge_i v_{i\varphi(i)} \leq \bigvee_\varphi \bigwedge_i u_{i\varphi(i)}$. Taking the join over the v gives $u \leq \bigvee_\varphi \bigwedge_i u_{i\varphi(i)}$. A family (f_{ij}) reduces to this case with $u = \bigwedge_i \bigvee_j f_{ij}$ and $u_{ij} = u \wedge f_{ij}$.

The same argument gives the exact content of (A) for $\mathrm{Sh}(L)$: (A) holds if and only if, for every u and every family of covers $\{u_{ik}\}_k$ of u, the element u is covered by elements v having, for each i, a disjoint decomposition subordinate to the i-th cover.

Proposition A.8. *The following are equivalent:* $\mathrm{Sh}(L)$ *satisfies (B); L is locally connected; L is* disjointly completely distributive*, that is, $b = \bigvee_\varphi \bigwedge_i f_{i\varphi(i)}$ whenever $b = \bigsqcup_j f_{ij}$ is a disjoint decomposition for each i. In particular every completely distributive frame is locally connected.*

Proof. Locally connected implies (B) by Proposition A.1. (B) implies disjoint complete distributivity by applying (B) to the subterminals: $y(b) = \prod_i \coprod_j y(f_{ij}) \cong \coprod_\varphi y(\bigwedge_i f_{i\varphi(i)})$.

Suppose L disjointly completely distributive, and fix b. Index all disjoint decompositions D of b (they form a set), and for a choice φ of a member $\varphi(D) \in D$ for each D put $e_\varphi = \bigwedge_D \varphi(D)$. By hypothesis $b = \bigvee_\varphi e_\varphi$, and $e_\varphi \wedge e_\psi = 0$ for $\varphi \neq \psi$, since they differ at some D and distinct

members of D are disjoint; so $b = \bigsqcup_\varphi e_\varphi$ is again a disjoint decomposition, the common refinement. Each nonzero e_φ is connected: if $e_\varphi = c_1 \sqcup c_2$, then $D' = \{c_1, c_2\} \cup \{e_\psi : \psi \neq \varphi\}$ is a disjoint decomposition of b, so $e_\varphi \leq \varphi(D')$; as e_φ is nonzero and disjoint from the e_ψ, $\psi \neq \varphi$, the member $\varphi(D')$ is c_1 or c_2, say c_1, and then $c_2 = c_2 \wedge e_\varphi \leq c_2 \wedge c_1 = 0$. So b is a join of connected elements, and L is locally connected. The last statement follows, since complete distributivity contains the disjoint case. □

Corollary A.9. *For a locale L the following are equivalent:* $\mathrm{Sh}(L)$ *satisfies (A) and (B); L is completely distributive;* $\mathrm{Sh}(L) \hookrightarrow \mathrm{PSh}(L)$ *is essential;* $\mathrm{Sh}(L)$ *is totally distributive. Moreover* $\mathrm{Sh}(L)$ *is a presheaf topos if and only if L is the frame of down-sets of a poset, that is, if and only if L is join-generated by completely join-prime elements.*

Proof. The subterminals form a small generating full subcategory of $\mathrm{Sh}(L)$, isomorphic to L, whose induced topology is the canonical one. Lemma A.7, Lemma A.4 and Theorem A.6 give the first equivalences.

If $\mathrm{Sh}(L) \simeq \widehat{C}$, the representables are connected projective objects. A projective P is a retract of the coproduct of subterminals $\coprod_{u,\, s \in P(u)} y(u)$, through a section of the canonical epimorphism; if P is connected the section lands in one summand, so P is a retract of a subterminal, hence subterminal, $P = y(p)$. Then p is completely join-prime: if $p \leq \bigvee S$, the epimorphism $\coprod_{s \in S} y(s \wedge p) \to y(p)$ splits, and by connectedness $p \leq s$ for one s. Every subterminal is a join of representables, so L is join-generated by completely join-prime elements. The converse is the standard fact that sheaves on the frame of down-sets of a poset Q are the presheaves on Q; it is not used below. □

So every completely distributive frame that is not of this form gives a counterexample to the question of Problem 1.

A.3. **The counterexample.** Let $X = [0, 1)$ with the open sets $[0, a)$, $0 \leq a \leq 1$. Its frame is $O(X) \cong [0, 1]$, a complete chain. (The same works for $\mathbf{R}$ with the open sets (a, ∞), whose frame is the chain $[-\infty, \infty]$.)

Sheaves. A sheaf on X is a functor $F\colon ((0, 1], \leq)^{\mathrm{op}} \to \mathbf{Set}$ with $F(a) \cong \lim_{0<b<a} F(b)$ for every a, together with $F(0) = 1$: a cover of $[0, a)$ not containing $[0, a)$ is cofinal in $\{[0, b) : b < a\}$, and the intersection of two members of a chain is the smaller one. Products are pointwise. Coproducts are pointwise on $(0, 1]$: a compatible family over a cofinal chain in a pointwise coproduct stays in a single summand, since restriction preserves the summand index, so the pointwise coproduct, with value 1 at 0, is already a sheaf.

(B) holds. At $a > 0$ both sides of (B) are $\coprod_\varphi \prod_i X_{i\varphi(i)}(a)$, by distributivity in $\mathbf{Set}$; at 0 both are 1. The cases where some J_i or I is empty are covered, since 0 is strict and empty products are 1. Equivalently, every nonzero element of a chain is connected ($b \wedge c = 0$ in a chain forces $b = 0$ or $c = 0$), so $\mathrm{Sh}(X)$ is locally connected and Proposition A.1 applies.

(A) holds. A complete chain is completely distributive: given $d < \bigwedge_i \bigvee_k a_{ik}$, choose for each i an index $\varphi(i)$ with $a_{i\varphi(i)} > d$; then $d \leq \bigwedge_i a_{i\varphi(i)}$, and the supremum over such d gives the inequality $\leq$. The direct argument of §A.2 then applies.

By Corollary A.9, $\mathrm{Sh}(X)$ is also an essential subtopos of a presheaf topos, and totally distributive. Directly: $\mathrm{Sh}(X) \subseteq \mathrm{PSh}((0, 1])$ is the subcategory of the functors continuous from below; the associated sheaf, $aF(a) = \lim_{b<a} F(b)$, is itself a limit, so it visibly preserves limits. In Kelly and Lawvere's terms the idempotent ideal is that of the strict inequalities $b < a$ [22], [23, 1.5.6].

Proposition A.10. $\mathrm{Sh}(X)$ *is not equivalent to a presheaf topos. It has no projective object other than* 0.

Proof. First argument: the frame $[0, 1]$ has no completely join-prime element, since 0 is the empty join and $x = \sup[0, x)$ for $x > 0$. Corollary A.9 concludes.

Second argument. Let P be projective, and let

$$e\colon \coprod_{(c,\, z \in P(c),\, c' < c)} y(c') \longrightarrow P$$

be given by $z|_{c'}$ on the summand (c, z, c'). It is an epimorphism, because $\sup\{c' : c' < c\} = c$. Choose a section σ of e, and suppose $x \in P(v)$ with $v > 0$. Since v is connected, $\sigma(x)$ lies in a single summand (c, z, c'), with $v \le c' < c$, and $e\sigma = \mathrm{id}$ gives $x = z|_v$. The section $\sigma(z)$ over c also lies in a single summand (c_1, z_1, c_1'), with $c \le c_1'$, and so does its restriction $\sigma(z)|_v = \sigma(z|_v) = \sigma(x)$. Hence $(c_1, z_1, c_1') = (c, z, c')$ and $c \le c' < c$, a contradiction. So $P(v) = \varnothing$ for all $v > 0$, that is, $P = 0$. □

Every nondegenerate presheaf topos has nonzero projectives, the representables, so either argument shows that the answer to Problem 1 is no.

An object is *connected* if it is not initial and is not a coproduct of two non-initial objects. For connected Q, the functor $\mathrm{Hom}(Q, -)$ preserves coproducts, as shown in the proof of Proposition A.1.

Proposition A.11. *For a Grothendieck topos E the following are equivalent: (1) E is equivalent to a presheaf topos $\widehat{C}$, C small; (2) E has a generating set of connected projective objects; (3) E satisfies (B) and has enough projectives. In particular the presheaf topoi are the topoi satisfying (A) and (B) with enough projectives, (A) being then automatic.*

Proof. (1)⇒(3): (B) holds by Proposition A.2, applied to the identity of $\widehat{C}$. A representable yc is projective, since $\mathrm{Hom}(yc, -)$ is evaluation at c and epimorphisms of $\widehat{C}$ are pointwise; so every X is covered by the projective $\coprod_{c,\, x \in X(c)} yc$.

(3)⇒(2): by Proposition A.1, E is locally connected, $\pi_0 \dashv \Delta$. Let P be projective, $\eta\colon P \to \Delta\pi_0 P = \coprod_{\pi_0 P} 1$ the unit, and P_k its fibre over $k \in \pi_0 P$, so that $P = \coprod_k P_k$. Decompositions of an object X into two summands are the maps $X \to 1 \sqcup 1 = \Delta 2$, and $\mathrm{Hom}(P, \Delta 2) \cong \mathrm{Hom}(\pi_0 P, 2)$ through $\chi \mapsto \Delta\chi \circ \eta$; so every summand of P is $\coprod_{k \in V} P_k$ for some $V \subseteq \pi_0 P$. A summand Q of P_k is a summand of P, hence of this form, and $P_l \subseteq Q \subseteq P_k$ forces $P_l = 0$ for $l \ne k$; so Q is 0 or P_k, and each nonzero P_k is connected. Each P_k is projective: given an epimorphism $e\colon Y \to Z$ and $q\colon P_k \to Z$, write $P = P_k \sqcup P'$ and lift $q \sqcup \mathrm{id}\colon P \to Z \sqcup P'$ along the epimorphism $e \sqcup \mathrm{id}$ to $l\colon P \to Y \sqcup P'$; the part of P_k that l sends into P' maps to $Z \times_{Z \sqcup P'} P' = 0$, so it is 0, and $l|_{P_k}$ lifts q through e. Now choose a generating set G, and for each $g \in G$ an epimorphism $P_g \to g$ with P_g projective; the nonzero components of the P_g form a generating set of connected projectives.

(2)⇒(1): let $C \subseteq E$ be the full subcategory on such a generating set. By the comparison lemma, $E \simeq \mathrm{Sh}(C, J)$, where a sieve S on c covers if and only if $\coprod_{(d,s) \in S} d \to c$ is an epimorphism. If it is, it has a section, since c is projective; the section factors through one summand (d, s), since c is connected, so $s\sigma = \mathrm{id}_c$ for some $\sigma\colon c \to d$, and $\mathrm{id}_c \in S$ because sieves are closed under composition. So only maximal sieves cover, and $\mathrm{Sh}(C, J) = \widehat{C}$. The last sentence follows with Theorem A.6. □

The example is locally connected and satisfies (A) and (B), so it has the infinitary exactness properties of **Set** that he lists. What fails is the projectivity of 1: $\Gamma = \mathrm{Hom}(1, -)$ does not preserve epimorphisms, which is not a commutation of limits with colimits. Whether $\mathrm{Sh}(X)$ has *every* exactness property of **Set** in his informal sense is not settled here, the notion not being defined precisely.

A.4. **Independence of (A) and (B).** *(A) without (B):* $\mathrm{Sh}(K)$*, K a Stone space such as the Cantor set.* For (A), every clopen v has, for any cover, a finite clopen refinement, which can be made disjoint; so the criterion of §A.2 holds with clopen elements v. Condition (B) fails because $\mathrm{Sh}(K)$ is not locally connected; explicitly, $\prod_n \Delta 2$ has the identity $K \to 2^{\mathbf{N}}$ as a global section, while the global sections of $\coprod_{2^{\mathbf{N}}} 1$ are the locally constant maps.

(B) without (A): $\mathrm{Sh}(\mathbf{R})$. It is locally connected, so (B) holds. Let $U_{n,k} = (k/2n,\, k/2n + 1/n)$, $k \in \mathbf{Z}$. Then $f_n\colon \coprod_k y(U_{n,k}) \to 1$ is an epimorphism for every n. A section of $\prod_n \coprod_k y(U_{n,k})$ over a connected open interval w forces $w \subseteq U_{n,k}$ for some k, for every n, so w is empty. Hence $\prod_n f_n$ has empty domain, and it is not an epimorphism.

(B) without (A), not localic: continuous $\mathbf{Z}_p$-sets. Orbits are connected, so (B) holds. Let $f_n\colon \mathbf{Z}_p/p^n\mathbf{Z}_p \to 1$. The product of the f_n is computed as the smooth part of the product of sets; its domain has no smooth points, since the stabiliser of every point is $\bigcap_n p^n\mathbf{Z}_p = 0$, which is not open. So the domain is empty, and $\prod_n f_n$ is not an epimorphism.

What is not addressed here: constructive versions without the axiom of choice; whether (B) with an abstract, non-canonical isomorphism suffices; and the precise meaning of « all exactness properties of **Set** ».

Appendix B. Problem three

This appendix proves the claims made under Problem 7. Statements that are complete only once the standing hypothesis (H) below holds are said to hold under (H).

B.1. **Definitions.** Fix $D = (D', D'')$, two sets of small diagram shapes. The 2-category $\Delta = \mathbf{Cat}_D$ has as objects the locally small categories with limits of type D' and colimits of type D'', as 1-cells the D-functors, which preserve both, and as 2-cells all natural transformations. Write $\mathrm{Hom}_D(R, C)$ for the category of D-functors $R \to C$.

Definition B.1 (Hypothesis (H))**.** His « problems one and two solved » (2:04:30): the following constructions exist in Δ, each defined by its 2-universal property against every $C \in \Delta$.

1. The envelope $j\colon I \to \widetilde{I}$ of a small category I: restriction $\mathrm{Hom}_D(\widetilde{I}, C) \to \mathrm{Hom}(I, C)$ is an equivalence.
2. The localisation $R \to R[\Sigma^{-1}]$ at a small set Σ of arrows: $\mathrm{Hom}_D(R[\Sigma^{-1}], C) \to \mathrm{Hom}_D(R, C)$ is fully faithful, with essential image the functors that invert Σ.
3. Equalising a small set of parallel pairs, defined in the same way.
4. Adjoining a small set H of arrows $h\colon a_h \to b_h$ between objects of R (his rule 3, 8:20:37): $R \to R\langle H\rangle$ with $\mathrm{Hom}_D(R\langle H\rangle, C) \simeq \{(\psi, \beta) : \psi \in \mathrm{Hom}_D(R, C),\ \beta_h\colon \psi a_h \to \psi b_h\}$, whose morphisms are the transformations $\psi \to \psi'$ compatible with the β's.
5. 2-colimits of small chains, and 2-pushouts (used only in §B.4).

Definition B.2 (Basic, algebraic)**.** Let I be a small discrete category and $i\colon I \to R$ a functor, with $R \in \Delta$. Then i is *basic* if there is a sequence $(R_\alpha)_{\alpha\le\gamma}$, indexed by a small ordinal γ, such that $R_0 = \widetilde{I}$; each $R_\alpha \to R_{\alpha+1}$ is one of the steps (2), (3) or (4), with small data; at a limit ordinal λ, R_λ is the 2-colimit in Δ of the R_β, $\beta < \lambda$; and there is an equivalence $R_\gamma \simeq R$ in Δ whose composite with $I \to \widetilde{I} \to R_\gamma$ is isomorphic to i. The category R is *algebraic* if some small discrete $I' \to R$ is basic.

This is the meaning of « basic » on his board at 2:00:40.

Definition B.3 (Condition (G) and saturation)**.** Condition (G) for i: every strictly full subcategory $R' \subseteq R$ that contains $i(I)$ and is closed under the D'-limits and the D''-colimits of R is all of R. The *saturation* of i: I_0 is the repletion of $i(I)$; $I_{\alpha+1}$ is the repletion of I_α together with the limits of the D'-diagrams in I_α and the colimits of the D''-diagrams in I_α; at limit ordinals, take unions.

B.2. **Saturation and the number of steps.**

Proposition B.4. *Condition (G) holds if and only if $\bigcup_{\alpha\in\mathrm{Ord}} I_\alpha = R$. If every shape in D has fewer than ρ arrows, for a regular cardinal ρ, then (G) holds if and only if $I_\rho = R$.*

This is the bound of [20, §3.5, p. 45].

Proof. The union $\bigcup_\alpha I_\alpha$ is replete and contains $i(I)$. It is closed under D-limits and D-colimits, because a diagram has a set of vertices and so lands in some I_α; and it is contained in every R' as in (G). For ρ: a diagram with fewer than ρ vertices in I_ρ lands in some I_α with $\alpha < \rho$, by regularity. □

For finitary D at most ω steps are needed. For $R = \mathbf{Set}$, $I = \{2\}$ and D all small limits, exactly two steps are needed: $I_2 = \mathbf{Set}$ by Proposition B.11 below, but $I_1 \neq \mathbf{Set}$, because no limit of a diagram of copies of 2 has three elements. Indeed every map $2 \to 2$ commutes with the ternary operation $x + y + z$ of $\mathbf{Z}/2$, so a limit of copies of 2, a set of compatible families, is closed under it; a three-element set $\{a, b, c\}$ closed under it has $a + b + c \in \{a, b, c\}$, say $a + b + c = a$, and then $b = c$. (This is `Buffalo.Topos.Part05.no_limit_of_bools` in the edition's Lean files [35].) Similarly, at the end of [20, §5.7, p. 100], $\mathbf{Cat}_0$ is the colimit-closure of $\{2\}$, but the idempotent category is not a colimit of a diagram of copies of 2, so one step does not suffice there either. For D with shapes of unbounded size we know no bound by a small ordinal; he expected one, without proof (8:08:14).

Remark B.5. Condition (G) implies that $\mathrm{Hom}_D(R, C) \to C^I$ is faithful: the objects X on which two transformations agree form a replete class closed under D-limits and D-colimits, since arrows into a preserved limit, or out of a preserved colimit, are determined by their components. So (G) lies strictly between « basic » (Theorem B.7) and faithfulness, which is too weak, as recalled after Problem 7.

B.3. **Basic implies (G), for every D.**

Lemma B.6. *The functor $i\colon I \to R$ satisfies (G) if and only if, for every fully faithful D-functor $u\colon B' \to R$ with $B' \in \Delta$ and every $i'\colon I \to B'$ with $ui' \cong i$, the functor u is an equivalence.*

Proof. If the condition holds, take $B' = R'$ and u the inclusion. R' has D-limits and D-colimits computed in R, so $R' \in \Delta$ and u is a D-functor; an equivalence onto a replete full subcategory forces $R' = R$. Conversely, the essential image of u is replete and contains $i(I)$. It is closed under D-limits: a D'-diagram on objects $u(b_k)$ lifts to B' because u is fully faithful, and u preserves its limit; colimits are dual. So by (G) the essential image is all of R. □

Theorem B.7. *Assume (H). If i is basic, (G) holds. This holds for every D, small or not, and even for a small I that is not discrete.*

Proof. We show by transfinite induction that (G) holds for $I \to R_\alpha$. At each stage let $u\colon B' \to R_\alpha$ be as in Lemma B.6, without loss of generality the inclusion of a replete full subcategory.

$\alpha = 0$. There is $i'\colon I \to B'$ with $ui' \cong j$. Extend i' to a D-functor $r\colon \widetilde{I} \to B'$ with $rj \cong i'$. Then $urj \cong j = \mathrm{id} \circ j$, and full faithfulness of restriction to I gives $ur \cong \mathrm{id}$. So u is essentially surjective.

Successor. Let $F\colon R_\alpha \to R_{\alpha+1}$ be the step. Then $F^{-1}(B')$ is replete, contains I, and is closed under D-limits and D-colimits, because F preserves them. By induction $F^{-1}(B') = R_\alpha$, so $F = uF'$ with F' a D-functor.

- Localisation: F' inverts Σ, because u is fully faithful and so reflects isomorphisms. So $F' \cong rF$ for some $r\colon R_{\alpha+1} \to B'$. Then $urF \cong F$, hence $ur \cong \mathrm{id}$.
- Equalising: the same, since u is faithful.
- Adjoining arrows: the universal arrows $\gamma_h\colon Fa_h \to Fb_h$ are $u(\beta'_h)$ for unique β'_h, because u is full. So (F', β') gives r with $(urF, ur\gamma) \cong (F, \gamma)$, and therefore $ur \cong \mathrm{id}$, by the equivalence $\mathrm{Hom}_D(R_{\alpha+1}, C) \simeq \{(\psi, \beta)\}$ with $C = R_{\alpha+1}$.

Limit ordinal. Each $F_\beta\colon R_\beta \to R_\lambda$ factors as uF'_β by induction. The coherence isomorphisms lift uniquely through the fully faithful u, so the F'_β form a pseudo-cocone and induce $r\colon R_\lambda \to B'$. As before, $ur \cong \mathrm{id}$.

End. Condition (G) is invariant under an equivalence compatible with I.

In every case the universal property is applied with target B', which lies in Δ. □

B.4. **Reduction of the converse to algebraicity.**

Lemma B.8. *Assume (H). Let x lie in the closure of $i(I)$ in R under D-limits and D-colimits. Then there are a basic $I \to S$, a D-functor $\Phi\colon S \to R$ with $\Phi|_I \cong i$, and an object $s \in S$ with $\Phi s \cong x$.*

Proof. By induction on the saturation stage of x. Let $x \cong \lim d$ with $d\colon J \to I_\beta$ a D'-diagram; the colimit case is dual.

(1) For each vertex d_k, choose a triple (S_k, Φ_k, s_k) by induction.
(2) Form the amalgamated 2-sum of the S_k over $\widetilde{I}$ (9:21:06). It is basic over I: the steps of each S_k, pushed out along $\widetilde{I} \to$ the sum, are again steps of the same kind.
(3) Adjoin the arrows of d, transported along $\Phi_k s_k \cong d_k$. They form a small set, because J is small.
(4) Equalise the composition and identity relations. They hold in R, so Φ extends to each new stage.
(5) There is now a diagram $\tilde{d}$ in the last stage S with $\Phi\tilde{d} \cong d$. Put $s = \lim \tilde{d}$; then $\Phi s \cong x$, since Φ preserves the limit. □

Theorem B.9. *Assume (H). For $i\colon I \to R$, the functor i is basic if and only if (G) holds and R is algebraic.*

Proof. One direction is Theorem B.7. Conversely, let $I' \to R$ be basic, through a sequence (R_α) from $\widetilde{I'}$. By (G) and Proposition B.4, every $x \in I'$ lies in the closure of I. Lemma B.8, applied once to the small family $(x)_{x\in I'}$ and amalgamating over $\widetilde{I}$ as in its step (2), gives (S, Φ) basic over I and objects $s_x \in S$ with isomorphisms $c_x\colon x \cong \Phi s_x$. Then build:

(1) $P = R \sqcup_{\widetilde{I'}} S$, the pseudo-pushout along $x \mapsto s_x$. It is basic over I: set $P_\alpha = R_\alpha \sqcup_{\widetilde{I'}} S$; then $P_0 \simeq S$, the pushout of a localisation, of an equalisation or of an adjunction of arrows is a step of the same kind, as one checks on the descriptions of $\mathrm{Hom}_D(-, C)$, and pushouts commute with colimits of chains.
(2) Make the two composites $S \rightrightarrows P$, the S-leg ι_S and $\iota_R\Phi$, isomorphic. Let $T_\beta\colon S_\beta \to S$, $\beta \le \delta$, come from the basic sequence of S over I. Let $M_\beta(C)$ be the category of pairs (χ, ε), with $\chi \in \mathrm{Hom}_D(P, C)$ and $\varepsilon\colon \chi\iota_S T_\beta \cong \chi\iota_R\Phi T_\beta$, whose morphisms are the transformations $\chi \to \chi'$ compatible with the ε's. We build from P, by small steps, $Q_\beta\colon P \to P'_\beta$ and a universal ε_β, that is, such that $\sigma \mapsto (\sigma Q_\beta, \sigma\varepsilon_\beta)$ is an equivalence $\mathrm{Hom}_D(P'_\beta, C) \simeq M_\beta(C)$ for every C, compatibly with restriction $M_{\beta'}(C) \to M_\beta(C)$ for $\beta \le \beta'$.
   - $\beta = 0$, $S_0 = \widetilde{I}$: adjoin arrows $e_k\colon \iota_S(k) \to \iota_R\Phi(k)$, $k \in I$, and their inverses, and equalise the inverse relations ($|I|$ data). This represents isomorphisms between the restrictions to I, which are the isomorphisms on $\widetilde{I}$, restriction $\mathrm{Hom}_D(\widetilde{I}, C) \to C^I$ being an equivalence.
   - Localisation or equalisation: restriction $\mathrm{Hom}_D(S_{\beta+1}, C) \to \mathrm{Hom}_D(S_\beta, C)$ is fully faithful, so $M_{\beta+1}(C) \to M_\beta(C)$ is an isomorphism, and $P'_{\beta+1} = P'_\beta$.
   - Adjoining a set H of arrows: by (H)(4), an isomorphism of D-functors on $S_{\beta+1}$ is an isomorphism of their restrictions to S_β compatible with the arrows $h \in H$. So $P'_{\beta+1}$ is P'_β with the two sides of the naturality square of ε_β at each h equalised ($|H|$ data).
   - Limit λ: let P'_λ be the 2-colimit of the P'_β. As S_λ is the 2-colimit of the S_β, an isomorphism of D-functors on S_λ is a compatible family of isomorphisms of their restrictions to the S_β, so $M_\lambda(C)$ is equivalent to the pseudo-limit of the $M_\beta(C)$, $\beta < \lambda$. So is $\mathrm{Hom}_D(P'_\lambda, C)$, of the equivalent chain $\mathrm{Hom}_D(P'_\beta, C)$; pseudo-limits of equivalent chains are equivalent, and ε_λ is the image of $\mathrm{id}_{P'_\lambda}$.

   Put $P' = P'_\delta$ and $e = \varepsilon_\delta$, an isomorphism $Q_\delta\iota_S \cong Q_\delta\iota_R\Phi$, since T_δ is an equivalence.
(3) For each $x \in I'$, equalise $e_{s_x} \circ \theta_x$ with the image of c_x, where θ is the isomorphism of the pushout ($|I'|$ data). Call the result P^*.

A D-functor $P^* \to C$ is a tuple $(\psi\colon R \to C,\ \varphi\colon S \to C,\ \theta,\ e)$ satisfying these equations. Such a tuple is determined by ψ up to unique isomorphism: one gets $\varphi \cong \psi\Phi$ through e, and θ is then forced. So restriction along the R-leg, $\mathrm{Hom}_D(P^*, C) \to \mathrm{Hom}_D(R, C)$, is an equivalence for every C. By the 2-Yoneda lemma, taking $C = R$ and $C = P^*$, the R-leg $R \to P^*$ is an

equivalence, compatible with I through the e_i. Every step used small data and the sequence has small length, so i is basic. □

Corollary B.10. *For a given Δ satisfying (H), Problem three has a positive answer if and only if every $R \in \Delta$ satisfying (G) for some small I is algebraic.*

B.5. **All small limits: the converse fails.**

Proposition B.11. *Every set X is an equaliser of two maps between powers of 2:*

$$X \cong \mathrm{Eq}(f, g\colon 2^{P(X)} \rightrightarrows 2^{P(P(X))\sqcup P(X)}),$$

$$f(u) = (S \mapsto u(\textstyle\bigcap S),\ A \mapsto u(A^c)), \qquad g(u) = (S \mapsto \textstyle\bigwedge_{A\in S} u(A),\ A \mapsto \neg u(A)).$$

Proof. The equaliser consists of the maps $u\colon P(X) \to 2$ preserving arbitrary intersections and complements, the complete Boolean homomorphisms. For $x \in X$, evaluation $u_x(A) = [x \in A]$ is one, and distinct points give distinct u_x. Conversely, let u be one, and $F = \{A : u(A) = 1\}$. Then $X = \bigcap \varnothing \in F$, so $\varnothing = X^c \notin F$; and F is closed under arbitrary intersections, so $\bigcap F \in F$ is nonempty. Pick $x \in \bigcap F$. If $A \in F$ then $x \in A$; if $x \in A$ then $x \notin A^c$, so $A^c \notin F$ and $A \in F$. Hence $u = u_x$, and x is unique since $\{x\} \in F$. □

Hence, in Δ the complete categories, $\{2\} \to \mathbf{Set}$ satisfies (G), in two steps. Dually, in Δ the cocomplete categories, $\{2\} \to \mathbf{Set}^{\mathrm{op}}$ satisfies (G).

Proposition B.12. *In $\mathbf{Set}^{\mathrm{op}}$ the object 2 is not λ-presentable for any regular cardinal λ.*

Proof. $\mathbf{Set}^{\mathrm{op}}(2, -)$ preserves λ-filtered colimits exactly when $P = \mathbf{Set}(-, 2)$ sends λ-cofiltered limits of $\mathbf{Set}$ to colimits. Take the chain $A_\alpha = [\alpha, \lambda)$, $\alpha < \lambda$, with limit $\bigcap_\alpha A_\alpha = \varnothing$. Then $P(\varnothing) = 1$, while $\varinjlim_\alpha P(A_\alpha)$ has at least two elements: the germs of $\varnothing$ and of A_α differ, since every A_β is nonempty. □

Proposition B.13. *Let Δ be the cocomplete categories with cocontinuous functors. If S is algebraic, every object of S is μ-presentable for some μ. Hence $\mathbf{Set}^{\mathrm{op}}$ is not algebraic, and $\{2\} \to \mathbf{Set}^{\mathrm{op}}$ satisfies (G) without being basic. Dually, $\{2\} \to \mathbf{Set}$ is a counterexample for D all small limits.*

This is the counterexample he pointed to (3:43:00, 3:44:08).

Proof. By induction along a basic sequence, $S \simeq \Phi\text{-Alg}$ for a small sketch (A, Φ), with $\mathrm{Cocts}[\Phi\text{-Alg}, B] \simeq \Phi\text{-Com}[A, B]$ for every cocomplete B, pseudo-naturally in B; at each step the sketch receives a functor from the previous one, along which restriction of comodels corresponds to restriction along the step. This is [20, Thm 6.11], by which Φ-Alg is the reflective orthogonality class of a regulus in $[A^{\mathrm{op}}, \mathbf{Set}]$ [20, Thm 6.5], together with [20, Thm 5.56]: with B cocomplete and A small, left adjoints may be replaced by cocontinuous functors. The steps:

- envelope: $(I, \varnothing)$, which gives $\mathbf{Set}^I$;
- enlarging the sketch: given a small set of objects, pass to the closure D_0 of $K(A)$ under λ-small colimits for λ large; it is small [20, §3.5], and $\Phi\text{-Alg} \simeq (D_0, \lambda\text{-colimits})\text{-Alg}$ with the same comodels in every cocomplete B [20, Prop. 6.21, Cor. 6.22, Thm 6.23];
- inverting $\sigma\colon X \to Y$: enlarge so that σ lies in D_0, then add the cylinder « Y, with leg σ, is the colimit of the one-object diagram X »;
- equalising: invert $Y \to \mathrm{coeq}$;
- adjoining arrows: adjoin the arrow freely to D_0, keeping the same cylinders;
- chains: at a limit λ, let A_λ be the pseudo-colimit in **Cat** of the sketches A_β, $\beta < \lambda$, small since λ and the A_β are, and Φ_λ the images of the cylinders of all the Φ_β. A functor $A_\lambda \to B$ is a compatible family of functors $A_\beta \to B$, and it is a comodel if and only if each of them is, so $\Phi_\lambda\text{-Com}[A_\lambda, B]$ is the pseudo-limit of the $\Phi_\beta\text{-Com}[A_\beta, B]$. By induction this chain is equivalent, pseudo-naturally in B, to the chain $\mathrm{Hom}_D(S_\beta, B)$, whose pseudo-limit is $\mathrm{Hom}_D(S_\lambda, B)$; and $\Phi_\lambda\text{-Com}[A_\lambda, B] \simeq \mathrm{Cocts}[\Phi_\lambda\text{-Alg}, B]$ by the

theorems just cited. By the 2-Yoneda lemma, B ranging over Δ, $S_\lambda \simeq \Phi_\lambda$-Alg, compatibly with the S_β.

In Φ-Alg the cylinders are cones over diagrams of presheaves of size $< \lambda$, so the orthogonality class is closed in $[A^{\mathrm{op}}, \mathbf{Set}]$ under λ-filtered colimits. Every presheaf on a small A is a small colimit of finitely presentable representables, hence μ-presentable for some $\mu \geq \lambda$; so every object of Φ-Alg is μ-presentable there. Proposition B.12 concludes. □

The fact behind this is that the algebraic categories are, up to duality, the locally presentable ones [13], [2, ch. 1]; the argument above uses only the results of [20] cited.

Remark B.14. Propositions B.11 and B.12 are checked in Lean, with the standard axioms only: `lean/Buffalo/ProblemThree.lean` in [35], with **Set** the types of a fixed universe. The file proves the bijection of every set with the equaliser of Proposition B.11; that the saturation of $\{2\}$ under limits of small diagrams has every set at its second stage and no three-element set at its first, so that (G) holds in exactly two steps; and that 2 is not κ-presentable in $\mathbf{Set}^{\mathrm{op}}$ for any regular κ, with Mathlib's notion of presentability and the chain of tails of κ. Proposition B.13 is not formalised: neither the notions « basic » and « algebraic » nor hypothesis (H) are modelled, nor Kelly's sketches, so the conclusions that $\mathbf{Set}^{\mathrm{op}}$ is not algebraic and that $\{2\} \to \mathbf{Set}$ is not basic rest on the argument above.

B.6. **Small D: the equivalence holds with the base fixed.**

Theorem B.15. *Let D be a small set of shapes, each with fewer than ρ arrows, ρ regular. Assume (H), with the constructions applied to small categories and small data giving small categories, as he assumes (2:08:14). Then for every small discrete I and every $i\colon I \to R$, the functor i is basic if and only if (G) holds. The base is fixed: I is not enlarged to* $\mathrm{Ob}\, R$.

First proof, by Theorem B.9. By Proposition B.4, $R = I_\rho$. Each I_α has a small set of isomorphism classes, since there is only a small set of diagrams of each of the small set of shapes [20, §3.5]; so R is essentially small. Its skeleton R_s is algebraic over $\mathrm{Ob}\, R_s$, as he shows (9:26:59): from $\widetilde{\mathrm{Ob}\, R_s}$, adjoin the arrows of R_s, equalise the composition relations, and invert the comparison maps for the small set of D-diagrams; by the 2-Yoneda lemma the result is equivalent to R. Theorem B.9 concludes. □

Second proof, direct. Build a chain S_β, $\beta \leq \rho$, with D-functors $\Phi_\beta\colon S_\beta \to R$ extending i. Each successor step (a) adjoins a formal arrow $a \to b$ for every arrow $\Phi a \to \Phi b$ of R, and (b) equalises every pair that Φ identifies; both are small, because S_β is small. Since ρ is regular and the shapes are ρ-small, the 2-colimit S_ρ can be computed in **Cat**: diagrams and parallel pairs come from earlier stages, and a functor out of S_ρ is a D-functor exactly when all its restrictions are. So Φ_ρ is fully faithful. Its essential image is closed under D-limits and D-colimits, since diagrams lift along a fully faithful functor; by (G) it is everything. A fully faithful, essentially surjective D-functor is an equivalence in Δ. □

Where (H) is available for small D: when $D' = \varnothing$, or dually $D'' = \varnothing$, [20, Thm 5.35] (the free $\mathcal{F}$-cocompletion is small) and [20, Thms 6.21–6.23] provide envelopes and localisations as small categories. For mixed small D, with D' and D'' both nonempty, we have not found a reference; he asserts it without proof (5:43:33). The cases of finite limits with I infinite but small, and of κ-small limits, fall under Theorem B.15: there is no counterexample with small D.

B.7. **The topos case.**

For topoi (D' the finite limits, D'' all small colimits, together with Giraud's exactness conditions and the existence of a generating set), the course on topoi states that every object of $R_{\alpha+1}$ is covered by images of objects of R_α (1:51:23), and, as a necessary condition, that if $I \to R$ is basic the finite products of the objects of I generate R (1:59:07). Both are false as stated.

- Take $R_\alpha = \mathbf{Set}$ and adjoin one arrow $1 \to 1 \sqcup 1$. This gives the classifying topos of decompositions of 1 into two summands, $R_{\alpha+1} = \mathbf{Set} \times \mathbf{Set}$, and the image of $\mathbf{Set}$ is the diagonal $\{(n,n)\}$. The object $(1,0)$ receives a map from (n,n) only when $n = 0$, so it is not covered.
- Take $I = \{x\}$. Start from the object classifier, invert $x \to 1$, which gives $\mathbf{Set}$, then adjoin $1 \to 1 \sqcup 1$. This makes $\{(1,1)\} \to \mathbf{Set} \times \mathbf{Set}$ basic, but the finite products of $(1,1)$ are just $(1,1)$, which does not generate.
- Condition (G) does hold here, as Theorem B.7 requires: pulling back $(\mathrm{inl}, \mathrm{inr}) \colon (1,1) \to (2,2)$ along inl gives $(1,0)$.
- An example of the same kind in $\mathbf{Z}/2$-sets: $Y = \mathbf{Z}/2 \sqcup 1$. There is no map $Y^n \to \mathbf{Z}/2$, yet $\mathbf{Z}/2 = Y \times_{Y/\tau} 1$, and the theory of torsors makes $\{Y\}$ basic.

The sufficient condition (2:00:13) is not affected. The repaired statement is « basic if and only if (G) ». Every topos is algebraic (1:06:58), so Theorem B.9 gives (G)⇒basic, its proof using only constructions available for topoi: classifying topoi, 2-pushouts (1:23:42) and subtopoi. For basic⇒(G), the proof of Theorem B.7 needs the subcategory B' to lie in Δ. By Lemma B.16 one may take for B' the closure of the image of I under finite limits and small colimits, which lies in Δ and is contained in every R' as in (G); the proof then shows that it is all of R_α at every stage, so (G) holds.

Lemma B.16. *Let R be a Grothendieck topos and K a small set of objects of R. The smallest replete full subcategory $B \subseteq R$ containing K and closed under finite limits and small colimits is a Grothendieck topos, and its inclusion preserves finite limits and small colimits.*

Proof. Write $R = \mathrm{Sh}(C, J)$ with C small, let $\mu \geq \aleph_0$ bound the number of arrows of C, and call a sheaf X *small* if $\sum_c |X(c)| < \kappa$, where $\kappa = (2^\theta)^+$ and $\theta \geq \mu$ is so large that the objects of K are small. Then $\nu^\mu < \kappa$ for every $\nu < \kappa$, and:

(a) finite limits of small sheaves are small, being computed pointwise;
(b) if F is a presheaf with fewer than κ elements, its sheafification F^{++} is small: $F^+(c)$ is a colimit over at most 2^μ sieves of sets $\mathrm{Match}(S, F)$ of cardinality at most ν^μ. Hence coproducts in R of fewer than κ small sheaves, and images of small sheaves, are small;
(c) every element $x \in X(c)$ of a sheaf lies in a small subsheaf, the image of $a(yc) \to X$, by (b);
(d) if the maps $s_l \to X$, $l \in L$, are jointly epimorphic and $W \subseteq X$ is small, then the images of the s_l for fewer than κ indices l already contain W: for each element w of W, choose a covering sieve on each member of which w lifts to some s_l, and one such l for each member.

Let S be the class of small objects of B, essentially a set, and B_S the class of $X \in R$ on which the maps from objects of S are jointly epimorphic. We show that $B \cap B_S$ contains K and is closed under small colimits and finite limits; then $B \subseteq B_S$. It contains $K \subseteq S$, and the terminal object, which lies in B and is small. A colimit in R of objects of $B \cap B_S$ lies in B, and the maps from S to it are jointly epimorphic, since the legs of a colimit are. Let $X \to Z \leftarrow Y$ be maps of $B \cap B_S$, and W a small subsheaf of $X \times_Z Y$; by (c) these cover it. Its images $W_X \subseteq X$ and $W_Y \subseteq Y$ are small by (b). By (d) and (b), there are $s_X \to X$ and $s_Y \to Y$ with $s_X, s_Y \in S$, coproducts of fewer than κ objects of S, whose images contain W_X and W_Y. Then $s_X \times_Z s_Y$ lies in B and is small by (a), so it lies in S, and its image in $X \times_Z Y$ is $\mathrm{im}(s_X) \times_Z \mathrm{im}(s_Y) \supseteq W$, epimorphisms being stable under pullback. So $X \times_Z Y \in B_S$.

Hence S is a small generating set of B. Finite limits and small colimits of B are computed in R, so coproducts in B are disjoint and universal, equivalence relations are effective and quotients universal. By Giraud's theorem B is a Grothendieck topos. □

B.8. **What remains open.** Large classes D beyond these cases: for instance $\mathbf{Cat}_D$ with D' the finite limits and D'' all colimits, without exactness conditions, where $\{2\} \to \mathbf{Set}^{\mathrm{op}}$ satisfies (G) and we do not know whether $\mathbf{Set}^{\mathrm{op}}$ is algebraic; mixed large classes, for which problem one itself may fail; and whether the number of saturation steps is bounded by a small ordinal

for unbounded D. By Corollary B.10, the question for each Δ is exactly whether (G) for some small I implies that R is algebraic.

Appendix C. Descent of quasi-coherent sheaves along universal effective epimorphisms

This appendix proves the claims made under Problem 13. In a category with fibre products, effective, regular and strict epimorphisms coincide; a universal effective epimorphism, a stable regular epimorphism in Mesablishvili's terms, is a single-arrow covering of the canonical topology. A family is universally effective-epimorphic if and only if its coproduct morphism is a universal effective epimorphism, coproducts of schemes being disjoint and universal. For $f\colon X \to Y$, write $\mathrm{Des}(f)$ for the category of quasi-coherent sheaves on X with descent data relative to f. In the vocabulary of the course, f is a *descent morphism* for quasi-coherent sheaves if $f^*\colon \mathrm{QCoh}(Y) \to \mathrm{Des}(f)$ is fully faithful, and an *effective descent morphism* if it is an equivalence.

C.1. **The affine case.**

Proposition C.1. *For a ring map $A \to B$ the following are equivalent:*

(a) $\operatorname{Spec} B \to \operatorname{Spec} A$ *is a universal effective epimorphism of schemes;*
(b) *it is a universal effective epimorphism of affine schemes;*
(c) *for every A-module M, the sequence $M \to M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B$ is an equaliser;*
(d) *$A \to B$ is pure, that is, universally injective;*
(e) *$A \to B$ is an effective descent morphism for modules;*
(f) *$A \to B$ is a descent morphism for modules.*

Proof. (a)⇒(b): affine schemes form a full subcategory of schemes closed under fibre products, so base changes and test objects can be restricted to affine ones. (b)⇔(c) is [38, 0EUA], applied to a one-member family; the proof of (b)⇒(c) uses the base change along $\operatorname{Spec}(A \oplus M)$ and the test object $\mathbf{A}^1$. (c)⇒(d) is clear. (d)⇔(e)⇔(f) is [38, 08XA], a theorem of Joyal and Tierney, of which [28] gives a published proof. (d)⇒(a): a pure morphism of affine schemes is a regular epimorphism in the category of schemes [26, Thm 5.7, Cor. 5.8], and this is stable under base change [26, Cor. 5.10]. □

So the affine universal effective epimorphisms are exactly the pure maps, whether one tests against affine schemes or all schemes. Mesablishvili proves more: a quasi-compact morphism of schemes is pure if and only if it is a stable regular epimorphism [26, Thm 5.11], and, in the paper to which that article's introduction refers [27], that a quasi-compact morphism is pure if and only if it is a stable effective descent morphism for quasi-coherent sheaves. Theorem C.4 below gives an independent proof of the direction used here.

C.2. **Positive results.**

Theorem C.2. *Every universal effective epimorphism $f\colon X \to Y$ of schemes is a descent morphism for quasi-coherent sheaves: $f^*\colon \mathrm{QCoh}(Y) \to \mathrm{Des}(f)$ is fully faithful. No quasi-compactness is needed.*

Proof.
(1) The question is Zariski-local on Y: sheaves of morphisms glue, and the base change of a universal effective epimorphism to an open subscheme of Y is again one. So assume $Y = \operatorname{Spec} A$.
(2) Let F be quasi-coherent on Y, let $Y' = \operatorname{Spec}(\mathcal{O}_Y \oplus F)$ be the square-zero extension, and $X' = X \times_Y Y'$.
(3) As sheaves on $|X| = |X'|$, $\mathcal{O}_{X'} = \mathcal{O}_X \oplus f^*F$, so $\Gamma(X', \mathcal{O}) = \Gamma(X, \mathcal{O}) \oplus \Gamma(X, f^*F)$, and likewise on $X' \times_{Y'} X'$. This needs no affineness of X.
(4) Test the effective epimorphism $X' \to Y'$ against $\mathbf{A}^1$, for which $\mathrm{Mor}(T, \mathbf{A}^1) = \Gamma(T, \mathcal{O}_T)$. Taking the F-components gives the exact sequence

$$\Gamma(Y, F) \to \Gamma(X, f^*F) \rightrightarrows \Gamma(X \times_Y X, f^*F).$$

(5) This sequence says that $\mathrm{Hom}_Y(\mathcal{O}, F) = \mathrm{Hom}_{\mathrm{Des}}(f^*\mathcal{O}, f^*F)$.
(6) Both $\mathrm{Hom}(-, F)$ and $\mathrm{Hom}_{\mathrm{Des}}(f^*-, f^*F)$ turn direct sums and cokernels into products and kernels, because f^* preserves colimits. A presentation $\mathcal{O}^{(I)} \to \mathcal{O}^{(J)} \to M \to 0$ then gives the result for every quasi-coherent M. □

Remark C.3. The affine case of Theorem C.2 is checked in Lean, with the standard axioms only: `lean/Buffalo/QcohDescent.lean` in [35]. If $N \to B \otimes_A N \rightrightarrows B \otimes_A B \otimes_A N$ is an equalizer, base change

$$\mathrm{Hom}_A(M, N) \to \mathrm{Hom}_{\mathrm{Des}}(B \otimes_A M,\, B \otimes_A N)$$

is bijective for every A-module M; conversely, bijectivity for $M = A$ gives the equalizer condition for N, and that condition for all modules is equivalent to $A \to B$ being pure [38, 08WE]. The formal proof needs no presentation of M: a B-linear map g is determined by $m \mapsto g(1 \otimes m)$, and it is compatible with the descent data exactly when this map lands in the equalizer. The step from universal effective epimorphisms to the equalizer condition [38, 0EUA], effectivity, and the non-affine case are not formalised.

Theorem C.4. *Every quasi-compact universal effective epimorphism $f\colon X \to Y$ of schemes is an effective descent morphism for quasi-coherent sheaves.*

Proof. (1) Reduction to $Y = \operatorname{Spec} A$. Once full faithfulness is known (Theorem C.2), effectivity is Zariski-local on Y: the objects descended over an open cover glue by uniqueness. This is his own argument in the algebraic geometry course.
(2) A finite affine cover. As X is quasi-compact, take a finite affine open cover (U_i) of X and put $U = \bigsqcup_i U_i = \operatorname{Spec} B$, with $q\colon U \to X$.
(3) The composite is pure. In the canonical topology, a sieve generated by a single arrow covers if and only if the arrow is a universal effective epimorphism; this is how he characterises the canonical topology (6:26:20). By transitivity of the topology, $g = f \circ q$ is a universal effective epimorphism, so $A \to B$ is pure by Proposition C.1.
(4) Descent along g. Given a descent datum (N, θ) on X relative to Y, the pulled-back datum $q^*(N, \theta)$ relative to g descends by [38, 08XA]: there are an A-module M and an isomorphism $\psi\colon g^*M \cong q^*N$ compatible with the descent data on $U \times_Y U$.
(5) Descent of ψ along q. Restrict ψ to $U \times_X U \subseteq U \times_Y U$. Along the diagonal $X \to X \times_Y X$, θ restricts to the identity (pull the cocycle condition back along the diagonal of $X \times_Y X \times_Y X$). Hence ψ is compatible with the canonical descent data for the Zariski cover q, and it descends to $\psi_0\colon f^*M \cong N$.
(6) Compatibility of ψ_0 with θ. It suffices to check it after pulling back along $U \times_Y U \to X \times_Y X$, which is a Zariski open covering (by the $U_i \times_Y U_j$), so that pulling back along it is faithful.
(7) Full faithfulness is Theorem C.2. □

Remark C.5. The proof gives more: if, locally on Y, some quasi-compact open $X_0 \subseteq X$ already has $X_0 \to Y$ a universal effective epimorphism, then f is an effective descent morphism. This is the statement of the course that a morphism through which an effective descent morphism factors is itself one.

The following lemma is not used later; it shows why a counterexample must be built from pieces that are not Mittag-Leffler, such as localisations, which is the shape of the example of §C.3.

Lemma C.6. *Let $(A \to B_j)_{j \in J}$ be a family such that $N \to \prod_j N \otimes_A B_j$ is injective for every A-module N. If every B_j is a Mittag-Leffler A-module, for instance finitely presented, then some finite subfamily is pure, that is, $N \to \prod_{j \in F} N \otimes_A B_j$ is injective for every N and some finite $F \subseteq J$.*

Proof. Otherwise, choose for every finite $F \subseteq J$ a module N_F and an element $n_F \neq 0$ of N_F killed in $N_F \otimes B_j$ for all $j \in F$. Choose an ultrafilter $\mathcal{U}$ on the finite subsets of J containing

each set $\{F : j \in F\}$, and let $N = \prod_{\mathcal{U}} N_F$ be the ultraproduct and $n = [n_F] \neq 0$. The Mittag-Leffler property makes $(\prod_F N_F) \otimes B_j \to \prod_F (N_F \otimes B_j)$ injective, so n dies in every $N \otimes B_j$, a contradiction. □

C.3. **The non-quasi-compact case.** *The family.* The following is [38, 0EUE]. Let $\mathcal{P}$ be the set of primes and U a nonprincipal ultrafilter on $\mathcal{P}$. For $A \in U$ let S_A be the multiplicative set generated by A and $\mathbf{Z}_A = S_A^{-1}\mathbf{Z}$, so that $\operatorname{Spec} \mathbf{Z}_A = \{(0)\} \cup A^c$; then $\mathbf{Z}_A \otimes \mathbf{Z}_B = \mathbf{Z}_{A \cup B}$, $\mathbf{Z}_{\{p\}^c} = \mathbf{Z}_{(p)}$ and $\mathbf{Z}_{\mathcal{P}} = \mathbf{Q}$. For every $\mathbf{Z}$-module M, M is the equaliser of $\prod_A M_A \rightrightarrows \prod_{A,B} M_{A \cup B}$ [38, 0EUF], and closed subsets glue. With [38, 0EUD], by which the equaliser condition and universal submersiveness give a universal effective epimorphism of schemes (on submersive morphisms see [37, §1]), it follows [38, 0EUG] that

$$\mathcal{W}_U = \{\operatorname{Spec} \mathbf{Z}_A \to \operatorname{Spec} \mathbf{Z}\}_{A \in U}, \quad \text{equivalently} \quad f \colon X = \coprod_{A \in U} \operatorname{Spec} \mathbf{Z}_A \to \operatorname{Spec} \mathbf{Z},$$

is a universal effective epimorphism of schemes. No finite subfamily is one, since every finite subfamily factors through some $\operatorname{Spec} \mathbf{Z}[1/p]$. So f is flat, surjective, not quasi-compact, and a covering for the canonical topology.

Descent data. Since $X \times_Y X = \coprod_{A,B} \operatorname{Spec} \mathbf{Z}_{A \cup B}$ and QCoh of a coproduct is the product of the categories of modules, a descent datum along f consists of $\mathbf{Z}_A$-modules N_A, $A \in U$, of isomorphisms $\varphi_{AB} \colon (N_A)_{A \cup B} \cong (N_B)_{A \cup B}$, and of the cocycle condition over $\mathbf{Z}_{A \cup B \cup C}$.

Lemma C.7. *Let U be an ultrafilter on a set α and $T \subseteq \alpha$. If $T \cap A^c$ is finite for every $A \in U$, then T is finite.*

Proof. If T is infinite, split it into two infinite halves $T = T_1 \sqcup T_2$. If $T_1^c \in U$, then $T \cap T_1 = T_1$ is infinite; otherwise $T_1 \in U$, so $T_2^c \supseteq T_1$ lies in U, and $T \cap T_2 = T_2$ is infinite. □

Proposition C.8. *In ZFC, descent data along f with all N_A torsion, or with all N_A torsion-free, are effective.*

Proof. Torsion data. If every N_A is torsion, then $N_A = \bigoplus_{p \notin A} T_p$ with $T_p = N_{\{p\}^c}[p^\infty]$, identified through the φ's, and $M = \bigoplus_p T_p$ descends the datum.

Torsion-free data. Put $V = N_{\mathcal{P}}$ and $N_{(p)} = N_{\{p\}^c} \subseteq V$, identified through the φ's. Over the principal ideal domain $\mathbf{Z}_A$, the torsion-free module N_A is $\bigcap_{p \notin A} N_{(p)}$. Put $M = \bigcap_p N_{(p)}$. Then $M \subseteq N_A$, so $M_A \subseteq N_A$. Fix $x \in V$ and let $T(x) = \{p : x \notin N_{(p)}\}$. For every $B \in U$, write $x = y/s$ with $y \in N_B$ and $s \in \mathbf{Z} \setminus 0$; then $T(x) \cap B^c \subseteq \{p : p \mid s\}$, which is finite. By Lemma C.7, $T(x)$ is finite. For $x \in N_A$ we have $T(x) \subseteq A$; for each $p \in T(x)$ some power $p^{k_p} x$ lies in $N_{(p)}$, so $s = \prod_{p \in T(x)} p^{k_p} \in S_A$ gives $sx \in M$. Hence $M_A = N_A$. □

Proposition C.9. *In ZFC, let $(T_p)_{p \in \mathcal{P}}$ be abelian groups, T_p being p-primary, for instance $T_p = \mathbf{Z}/p$, and $T = \bigoplus_p T_p$, so that $T_A = \bigoplus_{p \notin A} T_p$. Call a family $g = (g_A)_{A \in U}$, with $g_A \in \prod_{p \notin A} T_p$,* coherent *if $g_A - g_B$ is finitely supported on $A^c \cap B^c$, and* trivial *if there is $k \in \prod_p T_p$ with $g_A \equiv k$ on A^c up to finitely many primes, for every A. For coherent g, put $t_{AB} = (g_A - g_B)|_{A^c \cap B^c} \in T_{A \cup B}$, a 1-cocycle, and*

$$N_A = \mathbf{Z}_A \oplus T_A, \qquad \varphi_{AB}(r, t) = (r,\, t + r\, t_{AB}).$$

This is a descent datum, an extension of $\mathcal{O}$ by T, and it is effective if and only if g is trivial.

Proof. Suppose that (M, α_A) descends the datum. The projections $N_A \to \mathbf{Z}_A$ are compatible, so by [38, 0EUF] for $\mathbf{Z}$ they define $\pi \colon M \to \mathbf{Z}$. Then $\operatorname{coker}(\pi)_A = 0$ for all A, so $\operatorname{coker} \pi = 0$, again by [38, 0EUF]. Choose m with $\pi(m) = 1$. Then $\alpha_A(m) = (1, h_A)$ with $h_A \in T_A$, hence finitely supported. Compatibility gives $g_A + h_A = g_B + h_B$ on $A^c \cap B^c$. These glue to $k \in \prod_p T_p$, since every p lies in some A^c, for instance for $A = \{p\}^c$; and $g_A = k - h_A \equiv k$.

Conversely, if g is trivial through k, put $h_A = (k - g_A)|_{A^c} \in T_A$, $M = \mathbf{Z} \oplus T$ and $\alpha_A(r, t) = (r,\, t + r\, h_A)$. Then $\varphi_{AB} \alpha_A$ and α_B agree on $\mathbf{Z}_{A \cup B}$, because $h_A + t_{AB} = h_B$ on $A^c \cap B^c$. □

Proposition C.10. *In ZFC, f is an effective descent morphism for quasi-coherent sheaves if and only if, for every family (T_p) of p-primary groups, every coherent family in the sense of Proposition C.9 is trivial.*

Proof. Only if is Proposition C.9. Conversely, $\mathrm{Des}(f)$ is abelian, with kernels, cokernels and colimits computed componentwise, localisation being exact and commuting with colimits; f^* is exact, preserves colimits and is fully faithful (Theorem C.2). So the effective data, the essential image of f^*, are closed under cokernels of maps between them, finite direct sums and filtered colimits.

Let (N_A, φ) be a datum. Torsion submodules commute with localisation, so the $(N_A)_{\mathrm{tors}}$ form a torsion datum and the quotients a torsion-free one; by Proposition C.8 they are f^*M' and f^*M'', with $M' = \bigoplus_p T_p$ and T_p p-primary, and M'' torsion-free. Write M'' as the directed union of its finitely generated submodules F, which are free; the datum is the filtered colimit of its pullbacks to the f^*F, so we may assume $M'' = \mathbf{Z}^n$. For an extension $0 \to f^*M' \to N \to f^*\mathbf{Z}^n \to 0$, let E_m be its pullback along the m-th inclusion $f^*\mathbf{Z} \to f^*\mathbf{Z}^n$. The sum map $\bigoplus_m E_m \to N$ is onto, with kernel the $(t_m) \in (f^*M')^n$ with $\sum_m t_m = 0$, which is f^* of a submodule of M'^n; so N is effective once the E_m are.

So let $0 \to f^*M' \to E \to f^*\mathbf{Z} \to 0$. Choose $u_A \in E_A$ over $1 \in \mathbf{Z}_A$. Then $\varphi_{AB}(u_A) = u_B + t_{AB}$ with $t_{AB} \in M'_{A \cup B}$, and $t_{AC} = t_{AB} + t_{BC}$. Put $B_p = \{p\}^c \in U$, and for $p \notin A$ let $g_A(p)$ be $t_{AB_p} \in M'_{A \cup B_p} = T_p$. For $p \notin A \cup B$, the cocycle relation in $M'_{A \cup B \cup B_p} = T_p$ gives $g_A(p) - g_B(p) = t_{AB}(p)$; so g is coherent, $t_{AB} = (g_A - g_B)|_{A^c \cap B^c}$, and E is the datum of Proposition C.9 built from g. As g is trivial, E is effective. □

Topological translation. Let $\omega^* = \beta\mathcal{P} \setminus \mathcal{P}$, and view U as a point of ω^*. The sets $C \notin U$ correspond to the clopen sets $C^* \subseteq \omega^* \setminus \{U\}$. A coherent family of $\{0,1\}$-valued functions is the same thing as a clopen subset S of the open set $O = \omega^* \setminus \{U\}$, and the family is trivial if and only if S extends to a clopen subset of ω^*. A clopen S of O fails to extend if and only if U lies in the closure of both S and $O \setminus S$, that is, if and only if U is a strong butterfly point, with both wings clopen, in the sense of Fine and Gillman [11]; equivalently, the Stone–Čech compactification of $\omega^* \setminus \{U\}$ is not ω^*. A $\{0,1\}$-valued family that is not trivial as a family of sets stays nontrivial in $\prod_p \mathbf{Z}/p$: a trivialising k would give the trivialising set $W = \{p : k(p) = 1\}$.

Butterfly points of character $\aleph_1$. The butterfly lemma of Fine and Gillman [11] says that under the continuum hypothesis every point of ω^* is a strong butterfly point; see [31, Lemma 3.1], whose proof we have read. That proof uses the continuum hypothesis only to obtain a clopen neighbourhood base $\{U_\alpha : \alpha < \omega_1\}$, together with two properties of ω^* that hold in ZFC: it is an F-space, and nonempty G_δ sets have nonempty interior. So, in ZFC, every point of character $\aleph_1$ is a strong butterfly point. Van Douwen's problem P20 in [33] states this as known: a point with a local base of cardinality ω_1 has $\beta(\omega^* \setminus \{p\}) \neq \omega^*$.

Theorem C.11. *Suppose that some nonprincipal ultrafilter U on $\mathcal{P}$ is generated by $\aleph_1$ sets, as is every one under the continuum hypothesis. Then $f\colon \coprod_{A \in U} \operatorname{Spec} \mathbf{Z}_A \to \operatorname{Spec} \mathbf{Z}$ is a universal effective epimorphism of schemes* [38, 0EUG] *and a descent morphism for quasi-coherent sheaves (Theorem C.2), but not an effective descent morphism: the datum of Proposition C.9 built from a clopen of $\omega^* \setminus \{U\}$ that does not extend is not effective. Consequently* QCoh *is a separated prestack, but not a stack, for the canonical topology on schemes.*

Proof. The point U has character $\aleph_1$, so it is a strong butterfly point; the translation above gives a coherent $\{0,1\}$-valued family that is not trivial, hence not trivial in $\prod_p \mathbf{Z}/p$, and Proposition C.9 applies. □

What remains open. Van Douwen, Kunen and van Mill [9] give a model of Martin's axiom with $\mathfrak{c} = \aleph_2$ in which $\beta(\omega^* \setminus \{p\}) = \omega^*$ for every p. In that model every $\{0,1\}$-valued coherent family is trivial, so the obstruction used in Theorem C.11 vanishes for every U. Whether $\mathcal{W}_U$ is then effective we do not know: one would need triviality for all coefficient systems (T_p) (Proposition C.10), which C^*-embedding does not obviously give. Also open are whether some U gives effectivity in ZFC, and whether there is a counterexample in ZFC, from this family or another. The question appears to involve set theory in an essential way.

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